Assignment 1 Report

COSC 420 - Deep Learning

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Abstract

In this assignment, I implemented and trained a stable diffusion model designed to generate specific information using a large data set of images. The data set used, smallNORB, is a collection of images of toys with labels corresponding to the type of object, and viewing elevation among other things.

I developed a generative model which generates images that resemble the training images. In order to achieve this, I implemented a very basic version of Stable Diffusion utilizing the architecture of a U-Net to iteratively train the network to remove noise from an image. In doing so, it is possible to start with an image of pure noise and create a recognizable image by iteratively removing noise from the initial image. As this is still a novel technique for image generation and I was limited in computing power, the results were less than ideal but it still demonstrates the effectiveness of the model.

This report is a summary of the iterations I made when developing this model and the design choices I made along the way. I have also included my final results as well.

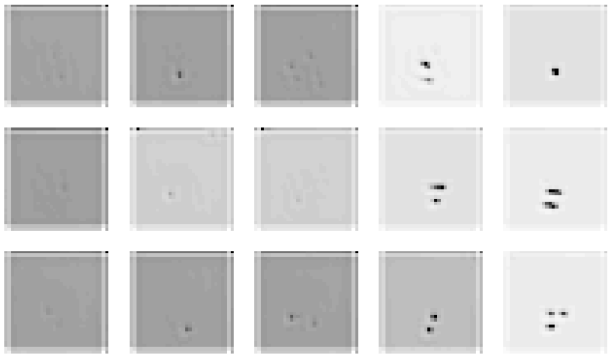
Generative Model

This generative model proved to be a great challenge since it is more novel and therefore less resources available to understand and implement it. However, I did find some great resources which helped explain the pipeline of the model as well as provide a guide on how to implement it. For this very basic introduction to the stable diffusion model using U-Net, I based my network off of a tutorial I found online from the journal Medium titled Make Diffusion model from scratch by the author under the name seachaos. Just so that I could get an understanding of the architecture and the process.

I first implemented this model without any layer normalization and using only 64 filters (instead of the recommended 128) for all the convolutions. I did this in an attempt to make the model as simplistic as possible. The results I got were promising but not ideal:

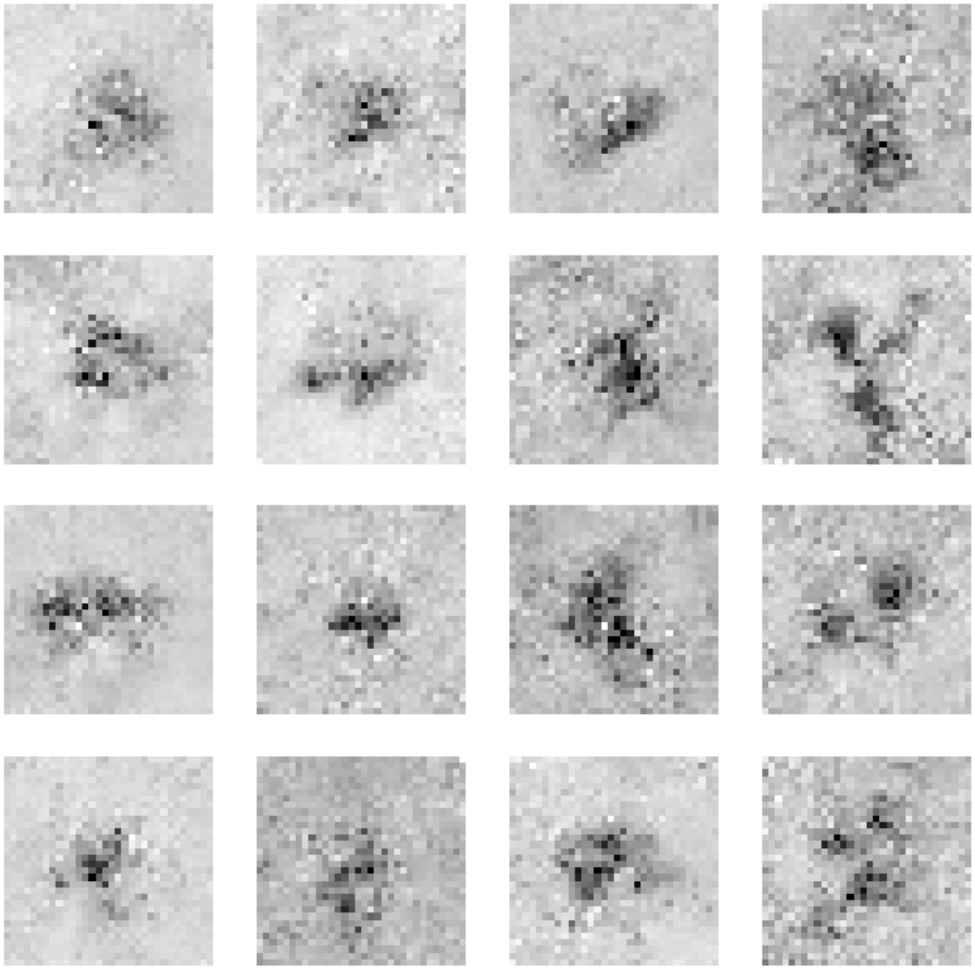
It is clear to see that there is obviously an object in the center of the frame but all of the images are too blurry to make anything out. This seemed to me to be an issue with overfitting and I also came to understand that the layer normalization was a crucial step in the model.

For my second iteration of the generative model, I attempted to fix the underfitting by using the recommended 128 filters for every convolution and I implemented the layer normalization prior to activations. The results of this model were much less than ideal.

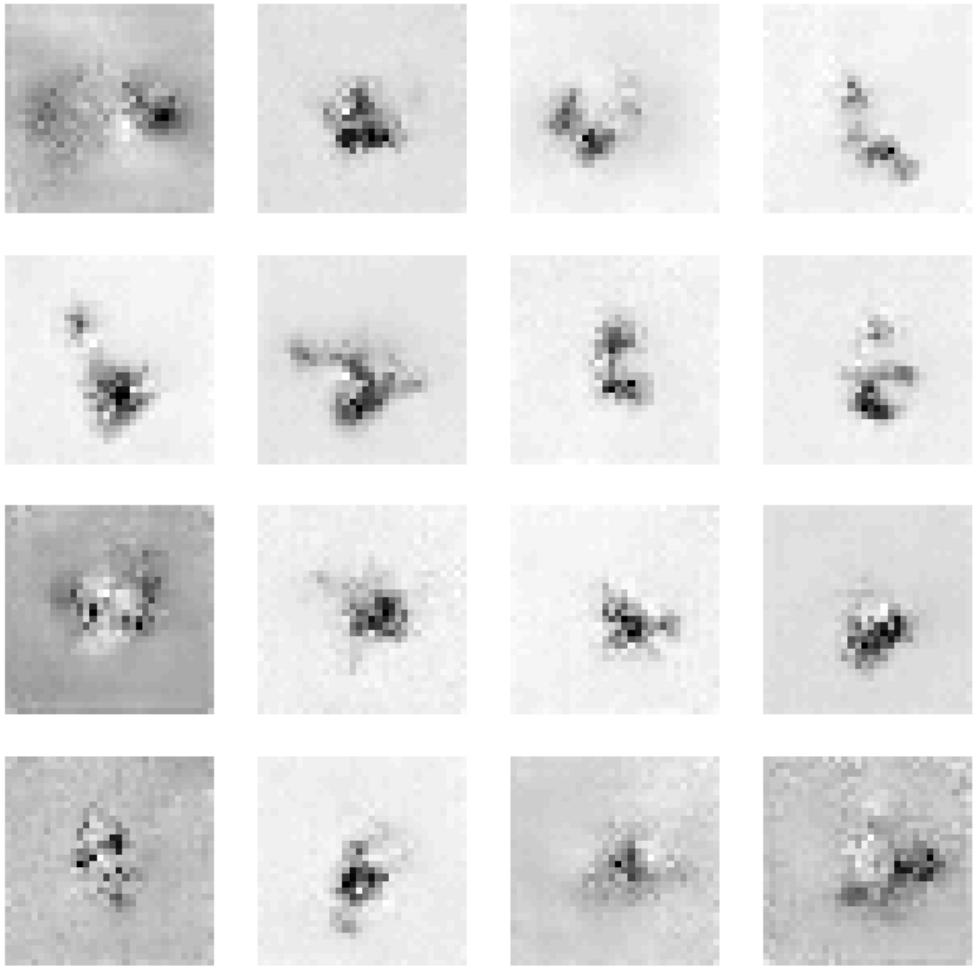


It seemed to be overfitting in a way and took way too long to train as well.

I then realized that I overlooked a very important step that seems to have a significant effect on training. I made sure to normalize the images before adding noise, so that all of the pixel values were between -1 and 1. After doing this, I saw very clear improvements from the network

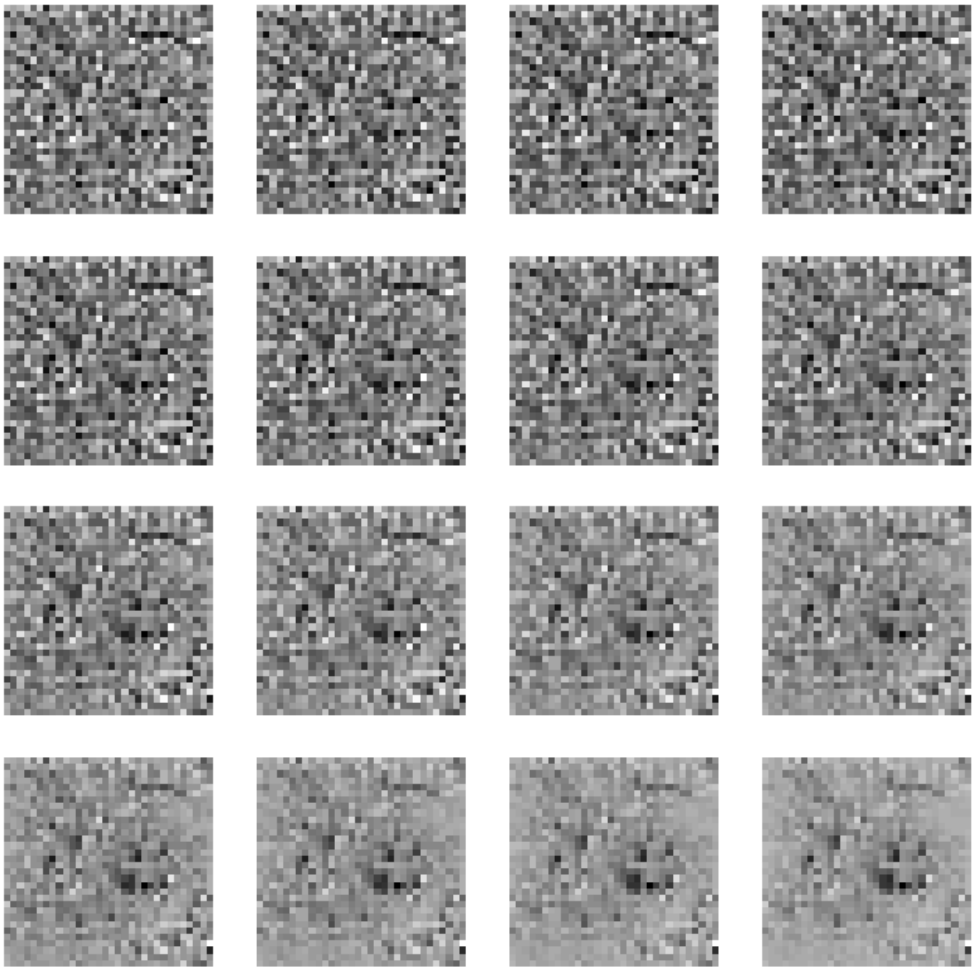


While still not perfect, it was definitely an improvement. I only trained it for 10 epochs but I noticed that the decrease in loss was not slowing when the 10th epoch was finished. Validation loss was still lower than the training loss and therefore I knew that the model could train for longer and would improve even more. I utilized the Checkpoints function keras offers to train for another 10 epochs. These are the results.



While far from perfect, these were the best results that I could get. If given more time and better computing power I would definitely train this algorithm for longer and use images of higher resolution to get better results.

I also configured a plot so that we can visualize the denoising process for one generative example



You can see the noise gradually disappear after each iteration.

Conclusion

I learned a lot from this exercise and feel a lot more confident about interpreting the results from training and knowing what sorts of things should be changed or added to improve the model. The generative model was really interesting to learn about and I think it would have been better if I had focused a little more time on polishing up the regression model rather than moving forward and committing much more time to developing the generative model.